AS-IS and TO-BE Process Analysis: Online Retail Order Fulfilment

1. Overview

The retailer's order fulfilment runs on manual stock checks, manual invoicing and manual courier handover, so customers get late updates and out-of-stock items are sometimes dispatched. This analysis maps the current (AS-IS) process, diagnoses five issues, and proposes an automated, system-driven TO-BE process with KPIs and risk controls.

- **Process boundary:** customer places order → order delivered to customer
- **Actors (swimlanes):** Customer, Website / Order System, Warehouse Team, Accounts Team, Courier Partner
- **Out of scope:** returns and refunds, procurement and supplier replenishment, marketing

2. AS-IS Process (Order Placement to Delivery)

Today stock is checked only after the order is accepted, and every hand-off — website to warehouse, warehouse to accounts, warehouse to courier — is done by email, phone or paper.

AS-IS: stock, invoice and courier steps all run by hand

Red markers link each step to issues 1 to 5 in Section 3

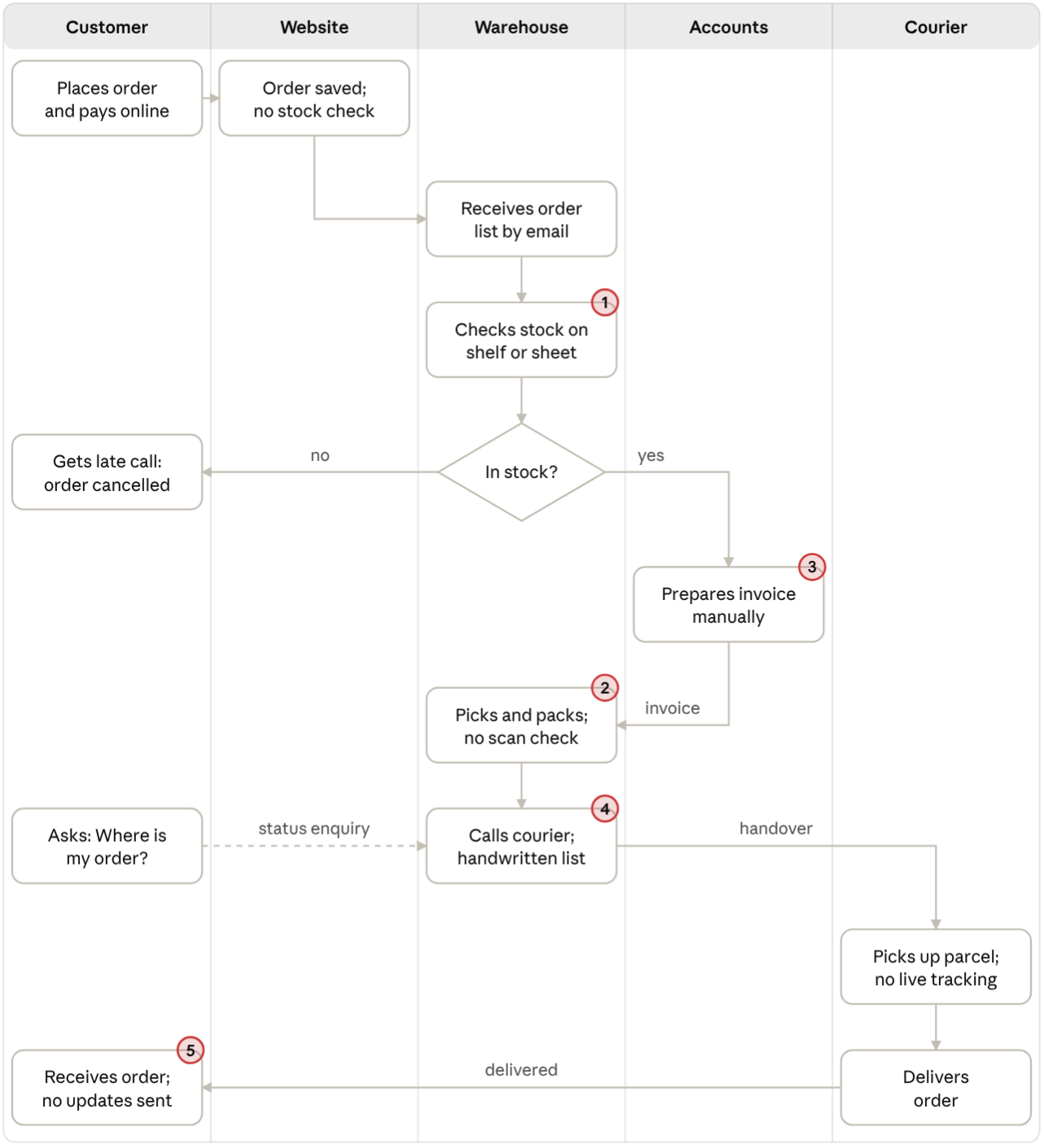


Figure 1 – AS-IS swimlane flowchart · order placement to delivery

The customer hears nothing between payment and delivery unless they chase the retailer; the dashed line shows those status enquiries landing on the warehouse team.

3. Process Issues and Business Impact

All five issues come from the same root: each hand-off depends on a person checking or re-typing data, with no shared real-time system. Issues 1 and 2 cause the out-of-stock dispatches; issues 3–5 cause the delays and late updates.

#	Issue	Root cause	Business impact
1	**Stock confirmed manually after the order is accepted**	Website stock is not linked to warehouse inventory; staff check shelves or a spreadsheet hours later	Overselling: orders are accepted for items not in stock, leading to cancellations, refunds and lost repeat customers
2	**No verification before dispatch**	Picking and packing are not scanned against the order; no final check	Wrong or out-of-stock items are dispatched, so the retailer pays for returns, re-shipping and refunds, and gets negative reviews
3	**Invoices prepared manually**	Accounts re-types order data into a billing tool and sends the invoice to the warehouse	Dispatch waits for the invoice (bottleneck); price/GST errors cause rework and compliance risk
4	**Manual courier handover**	Pickups booked by phone/email; handwritten manifest; tracking IDs entered later or not at all	Missed or late pickups, lost parcels with no proof of handover, longer delivery time
5	**Delayed, manual customer updates**	Customers are updated only when staff remember or are asked; no automatic status messages	High "Where is my order?" calls and emails, higher support cost, order cancellations and low customer satisfaction

4. TO-BE Process

In the TO-BE process an Order Management System (OMS) linked to warehouse inventory, invoicing and the courier's API replaces the manual hand-offs. Stock is checked before the order is accepted, and no parcel leaves without a scan match.

TO-BE: the system checks stock, invoices and books the courier

Green markers show where issues 1 to 5 from Section 3 are fixed

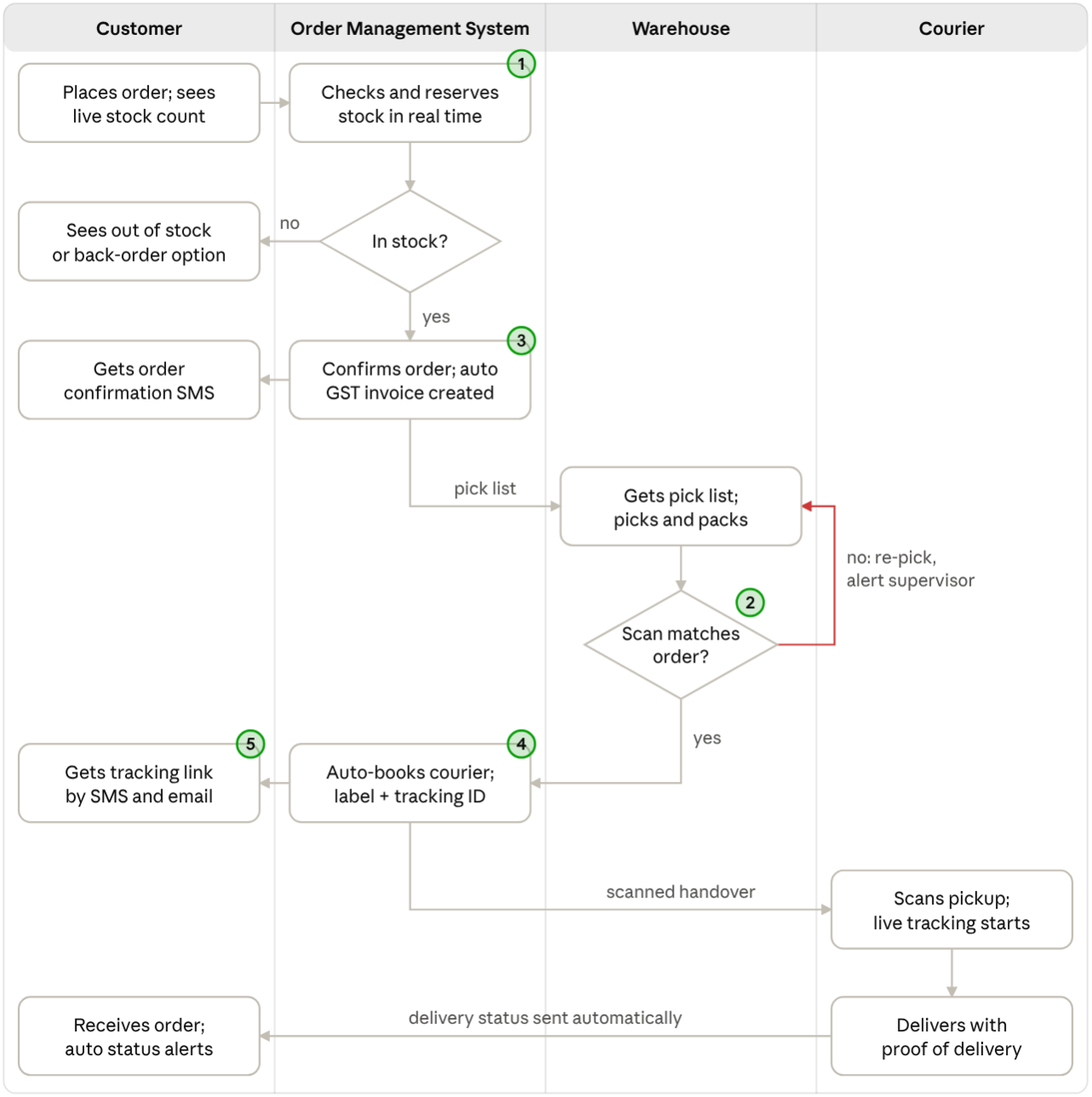


Figure 2 – TO-BE swimlane flowchart · order placement to delivery

The Accounts lane disappears from the daily flow: invoices are generated by the system, and Accounts only reviews exceptions and reconciles payments.

How the TO-BE process fixes each issue

Issue	TO-BE change	Result
1. Manual stock check	Website and warehouse share one real-time inventory; stock is reserved at checkout	Out-of-stock items cannot be ordered

Issue	TO-BE change	Result
2. No dispatch verification	Barcode scan at packing must match the order before the parcel is sealed	Wrong or missing items are caught before dispatch
3. Manual invoices	OMS creates the GST invoice automatically on order confirmation	No waiting for Accounts; no re-typing errors
4. Manual courier handover	OMS books the courier through its API and prints the shipping label with tracking ID; pickup is scanned	Pickups are booked on time, with proof of handover
5. Delayed customer updates	Automatic SMS/email at confirmation, dispatch (with tracking link) and delivery	Customers know the status without calling

5. Key Performance Indicators

Four KPIs track whether the TO-BE process fixes the five issues. Each is calculated automatically from system timestamps or reason codes, so no manual counting is needed. Targets are proposed; the AS-IS baseline should be measured for one month before go-live.

KPI	Definition and formula	How it is measured	Target
1. Order-to-Dispatch Time (Issues 3, 4)	Average hours from order confirmation to courier pickup scan. = Σ (pickup scan time − order confirmed time) ÷ orders dispatched	Timestamps from the order system and courier API; daily dashboard, weekly review	⩽ 24 hours for 95% of orders
2. Order Accuracy Rate (Issue 2)	% of orders delivered with the correct items and quantities. = (orders delivered − orders with wrong/missing-item complaints) ÷ orders delivered × 100	Pack-scan mismatch logs + support tickets and returns tagged "wrong/missing item"; weekly	⩾ 99.5%

KPI	Definition and formula	How it is measured	Target
3. Stock-out Cancellation Rate (Issue 1)	% of orders cancelled because the item was not in stock. = orders cancelled with reason "out of stock" ÷ total orders × 100	Mandatory cancellation reason code in the order system; weekly, by SKU	⩽ 0.5%
4. "Where Is My Order?" (WISMO) Contact Rate (Issue 5)	Order-status enquiries per 100 orders. = WISMO tickets, calls and chats ÷ total orders × 100	Helpdesk tickets tagged "order status"; monthly	50% lower than AS-IS baseline within 3 months

6. Process Risks and Recommended Controls

Automation removes manual errors but creates two new risks: wrong system stock data, and a failed system integration silently stopping orders.

Risk	Cause	Impact	Recommended controls
R1. Inventory data in the system does not match physical stock	Unscanned receipts, damaged goods not written off, returns not restocked in the system	Overselling continues even after automation; customers still receive cancellations	**Preventive:** every stock movement (receipt, return, damage) must be scanned; safety-stock buffer so the website shows "out of stock" at a set minimum quantity. **Detective:** weekly cycle counts of fast-moving SKUs, with inventory accuracy ⩾ 98% and any variance investigated

Risk	Cause	Impact	Recommended controls
R2. Integration failure between order system, invoicing and courier	API outage, wrong data mapping, courier system downtime	Orders stuck unpaid, uninvoiced or unshipped without anyone noticing; delayed deliveries	**Preventive:** test integrations before every release; backup manual process (documented) for invoice and courier booking. **Detective:** automated alert when an order stays in one status beyond its time limit (e.g., "Paid" > 4 hours, "Packed" > 12 hours); daily exception report reviewed by the operations lead